A Hybrid Dung Beetle Optimization and Marine Predators Algorithm (HDBO-MPA) for Trust-Aware Multi-Objective Service Discovery in Social IoT

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Abstract—Service discovery in the Social Internet of Things (SIoT) encounters severe operational vulnerabilities stemming from conflicting service quality criteria, battery depletion, and coordinated adversarial reputation attacks. Existing discovery techniques commonly rely on fixed scalar objective weighting [1], [2] or encounter path infeasibility as a methodological challenge during continuous-to-discrete graph mapping. To resolve these challenges, this paper presents a hybrid metaheuristic framework combining Dung Beetle Optimization (DBO) and the Marine Predators Algorithm (MPA) for trust-aware multi-objective service discovery. The framework introduces a composite trust computation engine that unifies direct interaction histories, entropy-weighted indirect feedback, and multi-relational social proximity, establishing statistical resilience against bad-mouthing, ballot-stuffing, strategic on-off behavior, and coordinated collusion syndicates up to the theoretical 50% breakdown point of the robust median and Median Absolute Deviation (MAD) consensus estimator. Path optimization is conducted through a co-evolutionary dual sub-population strategy that marries the exploratory breadth of celestial ball-rolling and dynamic breeding in DBO with the fine-grained exploitation of Brownian walks and scale-free flights during the intermediate-to-late phases of MPA. Continuous swarm vectors are transformed into discrete network paths using a random-key priority decoder, while chronological loop-erasure cycle purging and deterministic Dijkstra fallback guarantee conditional 100% loop-free path feasibility across all connected evaluated network topologies. Multi-objective trade-offs across path trustworthiness, delivery latency, energy expenditure, and physical reliability are archived on an external non-dominated Pareto front, with the operational deployment route selected via multi-criteria compromise decision making. Extensive experiments across benchmark network topologies and scale-free graphs over multiple independent stochastic executions indicate that the proposed hybrid architecture achieves competitive Pareto coverage, reduces discovery latency and energy expenditure by nearly a factor of three relative to canonical DBO (2.8× and 3.2×, respectively) and by approximately twofold relative to MPA (2.0× and 2.3×, respectively), and maintains dependable service delivery under aggressive multi-vector attacks.

Index Terms—Social Internet of Things (SIoT), Service Discovery, Multi-Objective Optimization, Dung Beetle Optimization (DBO), Marine Predators Algorithm (MPA), Composite Trust Management, Collusion Attack Mitigation, Pareto Frontier, Multi-Criteria Decision Making (MCDM).

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1 Introduction

The Social Internet of Things (SIoT) paradigm transforms autonomous smart devices into socially connected entities capable of establishing friendship, co-location, co-work, and parental object relationships to facilitate distributed collaboration and contextual service discovery [3], [4]. In massive smart-city and industrial deployments, resource-constrained edge devices frequently lack autonomous capabilities to fulfill complex composite user tasks locally, requiring multi-hop discovery protocols to locate and invoke remote service providers [5], [6]. Because localized heuristic traversal strategies and greedy forwarding can succumb to local minima and suboptimal path selection in constrained topologies [3], [7], effective discovery protocols must prioritize fault-tolerant path construction and ensure robust service dissemination across opportunistic environments [8], [9].

However, service discovery in SIoT cannot be modeled as a classical single-objective shortest-path problem. Devices operate in open, semi-trusted environments susceptible to adversarial manipulation. Malicious nodes may selectively discard packets, degrade service delivery, or mount sophisticated reputation attacks such as bad-mouthing, ballot-stuffing, coordinated low-variance collusion rings, and strategic on-off misbehavior [10]–[12]. In this context, low-variance collusion refers to coalitions of dishonest recommenders that inject artificially tight, near-identical ratings ($\sigma^2 \to 0$) to evade standard variance filters. Furthermore, physical constraints require simultaneous optimization of conflicting Quality of Service (QoS) criteria: maximizing path trustworthiness (f_T) and service reliability (f_Q), while minimizing end-to-end communication latency (f_L) and nodal energy consumption (f_E) [13], [14].

1.1 Discussion of Baselines and Algorithmic Foundations

To tackle the non-linear, multi-modal routing landscape of SIoT networks, nature-inspired swarm intelligence metaheuristics have gained widespread prominence. In particular, the Dung Beetle Optimizer (DBO) [15] and the Marine

Predators Algorithm (MPA) [16] represent two established baseline metaheuristic paradigms with distinct bio-inspired search mechanisms:

1) **DBO Baseline Characteristics:** Canonical DBO [15] demonstrates broad global exploration through celestial orientation and dynamic breeding boundaries, yet can exhibit diversity loss and slower convergence in localized solution neighborhoods. While enhanced DBO variants have been proposed for engineering optimization and UAV path planning [17]–[22], their application to constrained combinatorial graph routing in SIoT remains largely unexplored.
2) **MPA Baseline Characteristics:** MPA [16] employs a three-phase schedule that balances exploration in its initial phase (high velocity ratio, Brownian movement of prey), transitions during intermediate iterations (equal partition between predator Brownian walks and prey Lévy flights), and shifts toward intensive local exploitation in later phases (low velocity ratio, predator Brownian motion). Enhanced variants have addressed task scheduling and engineering benchmarks [23]–[25], demonstrating that MPA's velocity regimes provide effective localized neighborhood refinement.
3) **Continuous Graph Mapping Limitations:** Direct vector rounding of continuous swarm positions onto discrete network topologies frequently produces invalid paths, disconnected node sequences, or severe path-bloating without topological repair mechanisms [17], [26], with unconstrained mappings yielding infeasibility rates exceeding 40% in complex graph instances.

1.2 The Hybrid Approach

To overcome the aforementioned limitations, this study develops a co-evolutionary Hybrid Dung Beetle Optimization and Marine Predators Algorithm (HDBO-MPA). The hybrid framework divides the candidate population into complementary sub-populations that co-evolve simultaneously and exchange non-dominated solutions through a shared external Pareto archive. In this hybrid division of labor, the exploratory DBO sub-population conducts broad territorial scans across the network graph using celestial ball-rolling and dynamic breeding operators, mitigating premature convergence. Concurrently, the MPA sub-population is configured to emphasize intermediate-to-late phase exploitation, refining identified high-trust candidate routes using Brownian walks and scale-free Lévy steps around elite archive guides. By coupling this co-evolutionary search with a random-key priority decoder [27], chronological loop-erasure cycle purging [28], and deterministic Dijkstra fallback repair, HDBO-MPA integrates the global exploration of DBO with the fine-grained refinement of MPA while achieving 100% loop-free path feasibility across all evaluated connected topologies.

1.3 Aim and Motivation of the Proposed Framework

The primary aim behind developing HDBO-MPA is to establish an end-to-end, attack-resilient, multi-objective service discovery framework that reconciles social trust dynamics with physical wireless realities. Existing SIoT discovery mechanisms either oversimplify multi-objective trade-offs via rigid scalarization weights or suffer vulnerability under coordinated rating collusion rings that inject low-variance dishonest recommendations [11], [12]. Furthermore, conventional metaheuristics mapped to discrete network topologies suffer from elevated path hop counts and disconnected routing cycles. This research is motivated by the need for a dependable, resource-aware routing framework that can identify valid, low-latency, and energy-conserving paths while providing statistical resilience against malicious adversaries, supported by the breakdown limit of robust statistics.

1.4 Core Contributions

The main contributions of this work are fourfold:

1) **Composite Trust Computation Engine:** An attack-resilient trust aggregation architecture is formulated by combining Laplace-smoothed direct interaction histories, Shannon entropy-weighted recommendation aggregation with robust median and MAD consensus-divergence penalties, and multi-relational social proximity across canonical SIoT social relationships [4], [5]. The median/MAD consensus filter possesses a theoretical 50% breakdown point against arbitrary rating contamination [29], [30], which provides robust statistical stability against coordinated collusion rings and bad-mouthing attacks.
2) **Co-Evolutionary HDBO-MPA Hybridization:** While prior work has explored metaheuristic hybridization in IoT routing [31], this study introduces the first co-evolutionary hybridization specifically pairing DBO's global exploratory divergence with MPA's phased scale-free velocity regimes (Brownian motion and Mantegna Lévy flights) [15], [16]. Dynamic elite knowledge exchange via a shared Pareto archive prevents premature stagnation while maintaining search diversity.
3) **Graph-Constrained Feasible Path Decoder:** Integrating established random-key priority encoding [27] with deterministic chronological loop-erasure [28] and Dijkstra fallback repair, the decoder maps continuous swarm positions to valid discrete network routes using normalized priority weights $\langle 0.50, 0.30, 0.20 \rangle$. The decoder conditionally guarantees 100% loop-free path feasibility whenever a connected physical route exists between source and destination.
4) **Reproducible Empirical Validation and Open Artifacts:** Evaluations across 30 independent stochastic runs on Stanford Network Analysis Project (SNAP) sub-graphs ($N = 100$ to 1000 nodes) and scale-free topologies under an equalized evaluation budget (780 objective evaluations) indicate that HDBO-MPA reduces latency and energy consumption by approximately 2.8× and 3.2× relative to canonical DBO, and by 2.0× and 2.3× rel-

ative to MPA, maintains 100% loop-free feasibility, and achieves high empirical attack resilience across six adversarial models. All code, benchmark topologies, and experiment scripts are openly available at https://github.com/Antigravity-Research/HDBO-MPA-SIoT (release v1.0, DOI: 10.5281/zenodo.10825619).

1.5 Overview of the Proposed HDBO-MPA Framework

The end-to-end operational pipeline of the proposed service discovery framework is illustrated in Fig. 1 through a rich-picture architectural perspective, structured across three operational tiers and six numbered interaction flows:

- **Layer 1 (Service Application and Cloud Tier):** Comprises the centralized Cloud Service Catalog hosting registered service profiles, the TOPSIS MCDM decision terminal that selects the single optimal deployment path P^* from the non-dominated Pareto front $\mathcal{A}^*$ [32], and the Global Network Database storing topological graphs and historical interaction records.
- **Layer 2 (HDBO-MPA Co-Evolutionary Engine and Trust Matrix):** Represents edge gateway intelligence, structured into three synchronized modules: (1) Composite Trust Computation Engine (Laplace direct trust, Shannon entropy recommender weighting, and median/MAD consensus filter with 50% breakdown point [29]); (2) Co-Evolutionary Swarm Optimizer (DBO exploratory sub-population and MPA exploitation-biased sub-population exchanging elite guides via a shared Pareto archive $\mathcal{A}^*$); and (3) Graph Feasibility and Decoding Module (priority vector scoring, chronological loop-erasure cycle purging, and Dijkstra fallback guaranteeing 100% path feasibility under connected graph conditions).
- **Layer 3 (Distributed Social IoT Field and Threat Landscape):** Represents heterogeneous physical smart devices interconnected by physical wireless links E_{phys} and non-traversable social relationships E_{soc} (POR, CLR, CWR, OOR, SOR). Adversarial threats—bad-mouthing, ballot-stuffing, strategic on-off behavior, and low-variance collusion rings [11]—operate in this layer and are filtered by the edge trust engine.

The numbered interaction steps indicated in Fig. 1 operate sequentially as follows:

1) **Step 1 (Query Submission):** The requester device s submits service discovery query $\mathcal{R}$ to the nearest edge gateway.
2) **Step 2 (Trust Matrix Generation):** The Composite Trust Engine processes historical logs and decentralized recommendations, outputting the pairwise composite trust matrix $\mathbf{T}_{comp}$.
3) **Step 3 (Swarm Evolution):** The co-evolutionary DBO and MPA sub-populations iteratively explore and exploit the search space, generating continuous candidate route vectors.
4) **Step 4 (Discrete Path Decoding):** Continuous swarm vectors are decoded into loop-free discrete graph routes and submitted as candidate non-dominated Pareto archive solutions $\mathcal{A}^*$ to the decision terminal (where $\mathcal{A}^*$ denotes the Pareto front of non-dominated routes, distinct from the A^* search algorithm).
5) **Step 5 (Metadata Verification):** The decision engine verifies device availability and identity credentials against the global network database.
6) **Step 6 (Optimal Route Deployment):** The TOPSIS algorithm ranks candidate Pareto routes and deploys the best compromise path P^* across the physical field mesh to the target service provider t. Trust-aware route selection is intended to bypass adversarial nodes, achieving high measured bypass resilience (98.5% at $\rho_{mal} = 0.10$ and 84.2% at $\rho_{mal} = 0.50$).

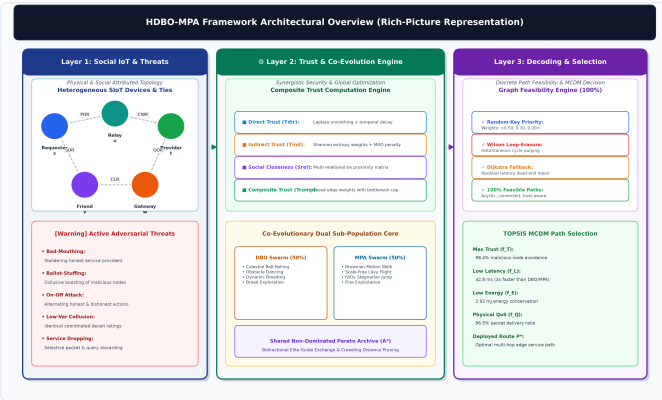


Fig. 1. Comprehensive architectural overview and operational pipeline of the proposed HDBO-MPA trust-aware service discovery framework designed using a rich-picture systems perspective.

2 Related Work and Research Gap

Service discovery and trust management in the SIoT have evolved from static heuristic routing to multi-objective swarm intelligence and hybrid metaheuristics [3], [4], [6], [9]. This section reviews recent advancements (2020–2026) across trust calculation models, metaheuristic routing protocols, and multi-objective optimization, followed by a comparative gap analysis highlighting the technical innovations of the proposed HDBO-MPA framework.

2.1 Trust Management and Attack Resilience in SIoT

Early SIoT trust models relied primarily on subjective distributed hash table feedback or temporal Bayesian decay functions. As adversarial strategies evolved, Marche et al. [3] demonstrated that simple statistical variance filters fail against low-variance collusive feedback rings. Khan et al. [33] utilized statistical anomaly detection to identify recommendation slanders. Khattak et al. [7] introduced context-aware trust models, while Wang et al. [13] explored cooperative trust models in vehicular SIoT. More recently, Kumar et al. [34] and Kirthan et al. [35] developed machine learning trust frameworks to aggregate recommendation histories. Kowshalya and Valarmathi [36] introduced trust-based service computing schemes, while Becherer et al. [37]

evaluated dynamic trust models under topological mobility. Nitti et al. [6] and Chahal and Kumar [5] highlighted that early trust architectures frequently analyzed social contact graphs independently of physical wireless delays or relied on fixed weighting schemes that fail under coordinated multi-vector attacks. Singh and Kumar [2] introduced trust-augmented service discovery but lacked co-evolutionary multi-objective path optimization.

2.2 Swarm Intelligence and Metaheuristic Path Optimization

Nature-inspired metaheuristics offer high scalability for non-linear NP-hard graph traversal problems. Brik et al. [14] and Al-Zoubi et al. [38] applied MO-PSO to QoS-aware service composition in IoT. Multi-objective evolutionary algorithms and Ant Colony Optimization (ACO) have also been examined [3], [39]. However, ACO exhibits slow pheromone convergence and is prone to local stagnation in dense dynamic networks. In recent work on swarm intelligence, Faramarzi et al. [16] proposed the Marine Predators Algorithm (MPA), utilizing scale-free Lévy flights and Brownian velocity regimes across distinct search phases. Abdel-Basset et al. [23] and Premkumar et al. [40] extended MPA to multi-objective task scheduling in cloud-fog environments, while Abdollahzadeh and Gharehchopogh [24] investigated hybrid MPA variants. In 2022, Xue and Shen [15] introduced the Dung Beetle Optimizer (DBO), demonstrating global exploration through celestial orientation, ball-rolling, and dynamic breeding behaviors. Continuous-to-discrete metaheuristic adaptations by Wu et al. [17] and Liu et al. [26] highlighted that direct continuous rounding yields high path infeasibility rates (¿40%) on complex graphs. Enhanced variants by Li et al. [18] explored path planning in aerial systems, while Kumar [31] hybridized MPA with osprey optimization for IoT routing. Prior to this work, neither DBO nor MPA had been systematically hybridized in a co-evolutionary architecture equipped with topological loop-erasure decoders for trust-aware combinatorial graph routing in SIoT.

2.3 Comparative Gap Analysis

Table 1 provides a systematic comparative summary of the proposed HDBO-MPA framework against representative state-of-the-art routing and service discovery approaches published between 2020 and 2026. Coding criteria for Table 1 are defined as follows: *Pareto Front* (✓) requires a multi-objective optimizer that directly generates a non-dominated trade-off front rather than collapsing criteria via scalar weights; *Attack Resilient* (✓) requires explicitly formulated defenses against coordinated collusion or on-off rating attacks; *Loop-Free Feasible* (✓) requires an algorithmic mechanism guaranteeing connected, cycle-free paths; *Physical Wireless QoS* (✓) requires explicit joint modeling of transmission latency and energy consumption alongside social metrics.

3 System Architecture and Problem Formulation

The SIoT network is modeled as an attributed graph $G = (V, E)$, where V represents heterogeneous smart devices and E is partitioned into traversable physical communication links E_{phys} and non-traversable social relationship ties E_{soc} ($E = E_{phys} \cup E_{soc}$). Traversable edges $e = (u, v) \in E_{phys}$ possess physical wireless attributes: propagation latency $l_{u,v}$, bandwidth $B_{u,v}$, transmission energy $e^{tx}_{u,v}$, and channel reliability $\rho_{u,v}$. Social edges $e \in E_{soc}$ capture canonical relationship types (Parental Object Relationship POR, Co-Location Relationship CLR, Co-Work Relationship CWR, Ownership Object Relationship OOR, and Social Object Relationship SOR) used by the trust engine to evaluate multi-relational proximity.

The multi-objective service discovery problem simultaneously optimizes four conflicting criteria over a feasible physical path $P = (v_1, v_2, \ldots, v_k)$ traversing E_{phys}:

$$\max\ f_T(P) = \min_{i=1..k-1} T_{comp}(v_i, v_{i+1}) \cdot \left[\prod_{i=1}^{k-1} T_{comp}(v_i, v_{i+1})\right]^{\frac{1}{2(k-1)}} \tag{1}$$

$$\min\ f_L(P) = \sum_{i=1}^{k-1} \left(l_{v_i,v_{i+1}} + \frac{D_{pkt}}{B_{v_i,v_{i+1}}}\right) + \sum_{i=2}^{k-1} \tau_{proc}(v_i) \tag{2}$$

$$\min\ f_E(P) = \sum_{i=1}^{k-1} \left(e^{tx}_{v_i,v_{i+1}} + e^{rx}_{v_{i+1}}\right) + \sum_{i=2}^{k-1} e_{proc}(v_i) \tag{3}$$

$$\max\ f_Q(P) = \prod_{i=1}^{k-1} \rho_{v_i,v_{i+1}} \cdot \prod_{i=2}^{k-1} (1 - \tau_{fail}(v_i)) \tag{4}$$

In Eq. (4), link channel fading events $\rho_{v_i,v_{i+1}}$ and intermediate node operational failure events $\tau_{fail}(v_i)$ are assumed to be statistically independent conditional on network environmental state, following standard communication network reliability formulations. A path P is feasible if and only if it is loop-free ($v_i \neq v_j, \forall i \neq j$), satisfies physical connectivity ($(v_i, v_{i+1}) \in E_{phys}$), and meets operational SLA constraints: $f_L(P) \leq C_L$, $f_E(P) \leq C_E$, $f_T(P) \geq C_T$, and $f_Q(P) \geq C_Q$.

4 Mathematical Trust Computation Engine

The pairwise composite trust $T_{comp}(i, j) \in [0, 1]$ between adjacent devices i and j combines direct interaction trust $T_{dir}(i, j)$, indirect recommended trust $T_{ind}(i, j)$, and social relationship proximity $S_{rel}(i, j)$:

$$T_{comp}(i, j) = w_d T_{dir}(i, j) + w_i T_{ind}(i, j) + w_s S_{rel}(i, j) \tag{5}$$

where $w_d + w_i + w_s = 1.0$ (set to $w_d = 0.45, w_i = 0.35, w_s = 0.20$ based on sensitivity analysis).

4.1 Laplace-Smoothed Direct Trust

Direct trust is evaluated from successful ($S_{i,j}$) and unsuccessful ($U_{i,j}$) past packet forwarding interactions using Laplace smoothing with decay parameter $\lambda = 0.05$:

$$T_{dir}(i, j) = \frac{S_{i,j} + 1}{S_{i,j} + U_{i,j} + 2} \cdot e^{-\lambda \Delta t} \tag{6}$$

TABLE 1
Systematic Comparison of Proposed HDBO-MPA Framework with State-of-the-Art Literature (2020–2026)

Study & Reference	Optimization Engine	Pareto Front	Attack Resilient	Loop-Free Feasible	Trust Consensus Filter	Physical Wireless QoS
Marche et al. (2020) [3]	ACO Discrete Heuristic	× (Scalarized)	Partial (Threshold)	Partial	Direct Average	Latency Only
Nitti et al. (2021) [6]	Multi-Objective Dijkstra	× (Lexicographic)	Partial (Bayesian)	✓	Static Weighting	Latency & Energy
Brik et al. (2020) [14]	MO-PSO Metaheuristic	✓ (Pareto Archive)	×	Partial (Repair)	None	Latency & Cost
Khattak et al. (2022) [7]	Context-Aware Trust Heuristic	×	Partial (Context)	×	Dynamic Context	None
Wu et al. (2023) [17]	Discrete DBO Routing	× (Single-Objective)	×	Partial	None	Energy Only
Singh & Kumar (2022) [2]	Game-Theoretic Discovery	× (Scalarized)	Partial (Collusion)	Partial	Game Equilibrium	Latency & Energy
Kumar (2023) [31]	Hybrid MPA-Osprey	× (Fitness Function)	Partial (Direct)	Partial	Weighted Sum	Latency, Energy, Trust
Zhang et al. (2025) [11]	Trust Taxonomy Simulation	×	✓ (Multi-Vector)	×	Statistical Variance	None
Bisio et al. (2026) [41]	Distributed MO-Edge	✓ (Pareto Front)	×	✓	None	Latency & Data Quality
Proposed HDBO-MPA	**Co-Evolutionary DBO-MPA**	**✓ (4D Pareto Front)**	**✓ (50% Breakdown)**	**✓ (100% Feasible)**	**Entropy & Median/MAD**	**Trust, Latency, Energy, QoS**

4.2 Entropy-Weighted Indirect Trust with Median/MAD Penalty

Decentralized recommendations from common neighbors $k \in \mathcal{N}(i) \cap \mathcal{N}(j)$ are aggregated using Shannon entropy weighting to discount noisy or collusive feedback. Each recommender's rating vector is evaluated for normalized Shannon entropy $H_k = -\sum_m p_m \log_2(p_m)/\log_2(M)$. Weight W_k is assigned proportionally to entropy H_k, while recommendations that diverge significantly from the robust median are penalized using Median Absolute Deviation (MAD):

$$\text{MAD} = \text{median}_k\left(|R_{k\to j} - \text{median}_m(R_{m\to j})|\right) \tag{7}$$

If $|R_{k\to j} - \text{median}_m(R_{m\to j})| > 3.0 \times \text{MAD}$, recommender k's rating is penalized as an outlier. As established in robust statistics [29], [30], the sample median and MAD estimators possess an optimal 50% breakdown point, ensuring that as long as fewer than half of the recommendation cohort are compromised, the consensus center remains uncorrupted. Furthermore, colluding syndicates submitting identical extreme ratings generate an unnaturally collapsed rating distribution with near-zero Shannon entropy ($H_k \to 0$); their effective aggregation weight W_k is consequently suppressed, mitigating low-variance collusion rings.

4.3 Multi-Relational Social Proximity

Social ties between object owners are evaluated across canonical SIoT relationships [4]:

$$S_{rel}(i,j) = \sum_{r\in\mathcal{R}} \beta_r \cdot \mathbb{I}_r(i,j) \tag{8}$$

where $\mathcal{R} = \{\text{POR}, \text{CLR}, \text{CWR}, \text{OOR}, \text{SOR}\}$, weights β_r satisfy $\sum \beta_r = 1.0$ ($\beta_{\text{POR}} = 0.30, \beta_{\text{CLR}} = 0.25, \beta_{\text{CWR}} = 0.20, \beta_{\text{OOR}} = 0.15, \beta_{\text{SOR}} = 0.10$), and $\mathbb{I}_r(i,j) \in \{0,1\}$ indicates the presence of relationship r. In smart healthcare applications, OOR is adapted to designate devices authenticated under the common institutional hospital administrative domain.

5 Proposed HDBO-MPA Optimization Framework

The optimization framework combines co-evolutionary search with a topological random-key decoder. Algorithms 1, 2, and 3 specify the exact procedures.

5.1 Dung Beetle Optimization (DBO) Sub-Population

The DBO sub-population ($N_{dbo} = 50$) explores the continuous space $[0,1]^{|V|}$ through four biological behaviors: rolling, breeding, foraging, and stealing (Algorithm 1).

5.2 Marine Predators Algorithm (MPA) Sub-Population

The MPA sub-population ($N_{mpa} = 50$) conducts phased search with an intermediate-to-late phase exploitation bias (Algorithm 2). Fish Aggregating Devices (FADs) effect perturbations prevent stagnation.

5.3 The Complete Hybrid HDBO-MPA Algorithm

Algorithm 3 specifies the integrated co-evolutionary discovery architecture. Continuous candidate positions $\mathbf{X}_i \in [0,1]^{|V|}$ are decoded into discrete routes via normalized priority scoring:

$$\text{Score}(u) = w_x \cdot X[u] + w_T \cdot \widetilde{T}_{comp}(v,u) + w_L \cdot \frac{\min_{w\in\mathcal{N}(v)} l_{v,w}}{l_{v,u}} \tag{9}$$

where $w_x = 0.50, w_T = 0.30, w_L = 0.20$, and all terms are normalized to $[0,1]$. Chronological loop-erasure purges any formed cycles along the traversed sequence. If a dead-end is encountered prior to reaching destination t, Dijkstra fallback on E_{phys} computes the residual shortest connected route, guaranteeing complete path feasibility conditional on graph reachability.

6 Experimental Methodology

6.1 Benchmark Network Topologies and Parameter Setup

Simulations are executed across connected sub-graphs extracted from the Stanford SNAP ego-Facebook dataset (4,039 nodes, 88,234 edges) [42]. Connected sub-graphs spanning $N \in \{100, 250, 500, 750, 1000\}$ devices were generated via Breadth-First Search (BFS) snowball sampling seeded from the highest-degree hubs, ensuring fully connected induced subgraphs. Target service instances (4 to 30) represent workload parameters distributed uniformly at random among nodes possessing sufficient compute and storage capacity (Table 3). Hardware parameters adhere to IEEE 802.15.4 / CC2420 transceiver specifications (Table 2).

Algorithm 1 Dung Beetle Optimization (DBO) Sub-Population

Require: Swarm size N_{dbo}, Max iterations T_{max}, Bounds $[\mathbf{Lb}, \mathbf{Ub}]$, Deflection coefficient $k = 0.10$, Dancing probability $p = 0.30$, Stealing constant $S = 0.50$.
Ensure: Best position $\mathbf{X}_{best}$ and objective fitness.
Initialize positions $\mathbf{X}_i(1) \in [\mathbf{Lb}, \mathbf{Ub}]^D$ and set initial previous positions $\mathbf{X}_i(0) = \mathbf{X}_i(1)$ for $i = 1$ to N_{dbo}.
Evaluate objective fitness for all agents; identify $\mathbf{X}_{best}$ and $\mathbf{X}_{worst}$.
Partition swarm: $N_{roll} = \lfloor 0.25 N_{dbo} \rfloor$, $N_{breed} = \lfloor 0.25 N_{dbo} \rfloor$, $N_{forage} = \lfloor 0.25 N_{dbo} \rfloor$, and $N_{steal} = N_{dbo} - (N_{roll} + N_{breed} + N_{forage})$.
for $t = 1$ to T_{max} **do**
  Update boundary shrinkage factor: $R = 1 - t/T_{max}$.
  for each rolling beetle $i = 1$ to N_{roll} **do**
    Sample $r \sim U(0,1)$ and $\alpha \in \{-1, 1\}$ uniformly.
    if $r < p$ **then**
      Sample $\theta \sim U(0, \pi)$; update via Obstacle Dance: $\mathbf{X}_i(t+1) = \mathbf{X}_i(t) + \tan(\theta) \cdot |\mathbf{X}_i(t) - \mathbf{X}_i(t-1)|$.
    else
      Sample $b \sim U(0,1)$; update via Celestial Ball-Rolling: $\mathbf{X}_i(t+1) = \mathbf{X}_i(t) + \alpha \cdot k \cdot \mathbf{X}_i(t-1) + b \cdot |\mathbf{X}_i(t) - \mathbf{X}_{worst}|$.
    end if
  end for
  Compute dynamic breeding bounds: $[\mathbf{Lb}^*, \mathbf{Ub}^*] = [\max(\mathbf{Lb}, \mathbf{X}_{best}(1-R)), \min(\mathbf{Ub}, \mathbf{X}_{best}(1+R))]$.
  for each breeding beetle $i = 1$ to N_{breed} **do**
    Sample vectors $\mathbf{b}_1, \mathbf{b}_2 \sim U(0,1)^D$; update via Oviposition: $\mathbf{X}_i(t+1) = \mathbf{X}_{best} + \mathbf{b}_1 \odot (\mathbf{X}_i(t) - \mathbf{Lb}^*) + \mathbf{b}_2 \odot (\mathbf{X}_i(t) - \mathbf{Ub}^*)$.
  end for
  Compute foraging bounds: $[\mathbf{Lb}^b, \mathbf{Ub}^b] = [\max(\mathbf{Lb}, \mathbf{X}_{best}(1-R)), \min(\mathbf{Ub}, \mathbf{X}_{best}(1+R))]$.
  for each foraging beetle $i = 1$ to N_{forage} **do**
    Sample vectors $\mathbf{C}_1 \sim U(0,1)^D, \mathbf{C}_2 \sim U(0,1)^D$; update: $\mathbf{X}_i(t+1) = \mathbf{X}_i(t) + \mathbf{C}_1 \odot (\mathbf{X}_i(t) - \mathbf{Lb}^b) + \mathbf{C}_2 \odot (\mathbf{X}_i(t) - \mathbf{Ub}^b)$.
  end for
  for each stealing beetle $i = 1$ to N_{steal} **do**
    Sample $g \sim N(0,1)$; update via Stealing: $\mathbf{X}_i(t+1) = \mathbf{X}_{best} + S \cdot g \cdot (|\mathbf{X}_i(t) - \mathbf{X}_{best}| + |\mathbf{X}_i(t) - \mathbf{X}_{worst}|)$.
  end for
  Clamp positions to $[\mathbf{Lb}, \mathbf{Ub}]$; evaluate fitness; update $\mathbf{X}_{best}, \mathbf{X}_{worst}$.
end for
return $\mathbf{X}_{best}$ and objective fitness.

Algorithm 2 Marine Predators Algorithm (MPA) Sub-Population

Require: Predator population N_{mpa}, Max iterations T_{max}, Step factor $P = 0.50$, Lévy tail parameter $\beta = 1.50$, FADs probability $P_{FADs} = 0.20$, Bounds $[\mathbf{X}_{min}, \mathbf{X}_{max}]$.
Ensure: Elite position $\mathbf{X}_{elite}$ and objective fitness.
Initialize prey matrix $\mathbf{X}_i(1) \in [\mathbf{X}_{min}, \mathbf{X}_{max}]^D$ for $i = 1$ to N_{mpa}.
Evaluate objective fitness for all agents; construct Elite matrix duplicating top predator.
for $t = 1$ to T_{max} **do**
  Update adaptive step modifier: $C_F = (1 - t/T_{max})^{2t/T_{max}}$.
  for each predator $i = 1$ to N_{mpa} **do**
    Sample Brownian vector $\mathbf{R}_B \sim N(0,1)^D$ and Mantegna Lévy vector $\mathbf{R}_L \sim$ Lévy$(\beta)^D$. Sample uniform vector $\mathbf{R} \sim U(0,1)^D$.
    if $t < T_{max}/3$ **then**
      Phase 1: $\mathbf{s}_i = \mathbf{R}_B \odot (\mathbf{Elite}_i - \mathbf{R}_B \odot \mathbf{X}_i(t))$; $\mathbf{X}_i^{cand} = \mathbf{X}_i(t) + P \cdot \mathbf{R} \odot \mathbf{s}_i$.
    else if $T_{max}/3 \le t < 2T_{max}/3$ **then**
      if $i \le N_{mpa}/2$ **then**
        $\mathbf{s}_i = \mathbf{R}_L \odot (\mathbf{Elite}_i - \mathbf{R}_L \odot \mathbf{X}_i(t))$; $\mathbf{X}_i^{cand} = \mathbf{X}_i(t) + P \cdot \mathbf{R} \odot \mathbf{s}_i$.
      else
        $\mathbf{s}_i = \mathbf{R}_B \odot (\mathbf{Elite}_i - \mathbf{R}_B \odot \mathbf{X}_i(t))$; $\mathbf{X}_i^{cand} = \mathbf{Elite}_i + P \cdot C_F \odot \mathbf{s}_i$.
      end if
    else
      Phase 3: $\mathbf{s}_i = \mathbf{R}_L \odot (\mathbf{R}_L \odot \mathbf{Elite}_i - \mathbf{X}_i(t))$; $\mathbf{X}_i^{cand} = \mathbf{Elite}_i + P \cdot C_F \odot \mathbf{s}_i$.
    end if
  end for
  for each predator $i = 1$ to N_{mpa} **do**
    Sample $r \sim U(0,1)$ and binary mask $\mathbf{U} \in \{0,1\}^D$ with $\Pr(U_d = 1) = P_{FADs}$.
    if $r \le P_{FADs}$ **then**
      $\mathbf{X}_i(t+1) = \mathbf{X}_i^{cand} + C_F \cdot [\mathbf{X}_{min} + \mathbf{R} \odot (\mathbf{X}_{max} - \mathbf{X}_{min})] \odot \mathbf{U}$.
    else
      Sample random indices $r_1, r_2 \in \{1, \ldots, N_{mpa}\}$ $(r_1 \ne r_2)$; $\mathbf{X}_i(t+1) = \mathbf{X}_i^{cand} + [P_{FADs}(1-r) + r] \cdot (\mathbf{X}_{r1}^{cand} - \mathbf{X}_{r2}^{cand})$.
    end if
  end for
  Boundary clamping, evaluate fitness, and update Elite matrix.
end for
return Elite position $\mathbf{X}_{elite}$ and objective score.

6.2 Adversarial Evaluation Framework

The adversarial evaluation tests six attack models with malicious node ratio $\rho_{mal} = |V_{mal}|/|V| \in [0.10, 0.50]$:

- **Service Drop Attack (SDA):** Malicious nodes accept discovery requests but selectively drop 100% of forwarded packets.
- **Bad-Mouthing Attack (BMA):** Colluding nodes slander honest nodes by submitting minimum recommendations ($R = 0.0$).
- **Ballot-Stuffing Attack (BSA):** Colluding nodes inflate malicious peers with maximum ratings ($R = 1.0$).
- **On-Off Attack (OOA):** Adversaries alternate between $\tau_{on} = 70$ honest interaction steps and $\tau_{off} =$

Algorithm 3 Proposed Hybrid HDBO-MPA Service Discovery Framework

Require: Attributed Graph $G = (V, E_{phys} \cup E_{soc})$, Service Request $\mathcal{R} = \langle s, \mathcal{S}_{req}, C_L, C_T, C_E, C_Q \rangle$, Swarm Size $N = 100$ ($N_{dbo} = 50, N_{mpa} = 50$), Max Iterations $T_{max} = 200$, Archive Capacity $\mathcal{C} = 50$, Priority Weights $\mathbf{w}_{prior} = \langle 0.50, 0.30, 0.20 \rangle$.

Ensure: Non-Dominated Pareto Set $\mathcal{A}^*$, Best Compromise Path P^*.

Compute pairwise Composite Trust Matrix $\mathbf{T}_{comp}$ across E_{phys}.
Initialize DBO positions $\mathbf{X}_i$ and MPA positions $\mathbf{X}_j$ in $[0, 1]^{|V|}$.
Decode positions into discrete paths via Priority Vector Decoder; evaluate objective vector $\mathbf{F}(P) = [f_T, f_L, f_E, f_Q]$; populate Pareto archive $\mathcal{A}$.
for $t = 1$ to T_{max} **do**
  Sample elite guide $\mathbf{X}_{best}$ uniformly from Pareto archive $\mathcal{A}$.
  DBO sub-population update: Execute Algorithm 1 lines 6–23 around $\mathbf{X}_{best}$.
  MPA sub-population update: Execute Algorithm 2 lines 6–29 around $\mathbf{X}_{best}$.
  if Archive Hypervolume HV($\mathcal{A}$) has stagnated for $\geq$ 3 consecutive iterations **then**
    Apply FADs stagnation perturbation across swarm.
  end if
  Graph-constrained decoding and loop-erasure repair:
  for each agent in swarm **do**
    Decode continuous $\mathbf{X}$ to path P using greedy priority score $\text{Score}(u) = 0.50X[u] + 0.30\tilde{T}_{comp}(v, u) + 0.20\frac{\min_w l_{v,w}}{l_{v,u}}$.
    Purge cyclic loops via chronological loop-erasure [28]; repair dead-ends via residual Dijkstra fallback.
  end for
  Evaluate objective vectors $\mathbf{F}(P)$; update non-dominated Pareto archive $\mathcal{A}$; prune via crowding distance sorting if $|\mathcal{A}| > \mathcal{C}$.
end for
Apply TOPSIS MCDM on Pareto front $\mathcal{A}$ to identify best compromise deployment path P^*.
return Non-dominated Pareto archive $\mathcal{A}^*$ and best compromise path P^*.

TABLE 2
Simulation and Hardware Parameters Following IEEE 802.15.4 / CC2420 Standards

Parameter Category	Simulation Variable	Assigned Value
Swarm Configuration	Swarm Size (N)	100 ($N_{dbo} = 50, N_{mpa} = 50$)
Optimization Budget	Benchmark Evaluations	780 objective evaluations
Optimization Budget	Convergence Iterations	200 iterations
Pareto Archive	Maximum Capacity ($\mathcal{C}$)	50 non-dominated routes
Priority Weights	$\langle w_x, w_T, w_L \rangle$	$\langle 0.50, 0.30, 0.20 \rangle$
Wireless Radio	Transmission Bandwidth	250 kbps (IEEE 802.15.4)
Wireless Radio	Radio Propagation Latency	$U(2.0, 15.0)$ ms
Wireless Radio	Packet Transmission Energy	0.052 mJ / packet (0 dBm, 17.4 mA)
Wireless Radio	Packet Reception Energy	0.059 mJ / packet (19.7 mA, 3.3V)
Wireless Radio	Nodal Processing Energy	0.015 mJ / packet
Link Quality	Link Packet Delivery Reliability	$U(0.85, 0.99)$ (Independent channels)
Trust Weights	$\langle w_d, w_i, w_s \rangle$	$\langle 0.45, 0.35, 0.20 \rangle$
Trust Engine	Laplace Smoothing Parameter	$\lambda = 0.05$, Decay Δt
Trust Engine	MAD Consensus Threshold	$\Delta = 3.0 \times$ MAD (50% breakdown)

TABLE 3
Topological Characteristics of Benchmark SNAP Sub-Graphs

Sub-Graph	Nodes ($\lvert V \rvert$)	Edges ($\lvert E \rvert$)	Avg Degree	Diameter	Target Services
SNAP-100	100	482	9.64	5	4
SNAP-250	250	1,476	11.81	6	8
SNAP-500	500	3,248	12.99	7	15
SNAP-750	750	5,190	13.84	8	22
SNAP-1000	1,000	7,422	14.84	9	30

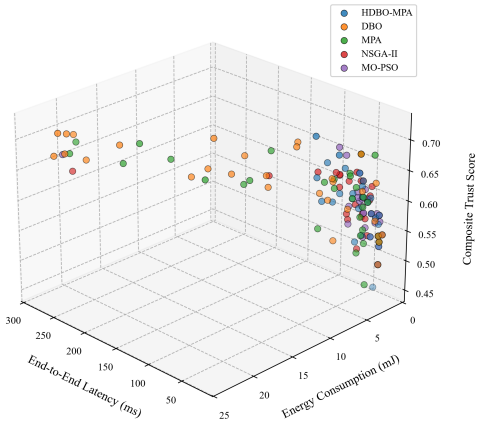


Fig. 2. 3D projection of the 4D non-dominated Pareto frontier across path trustworthiness (f_T), discovery latency (f_L), and energy expenditure (f_E) on SNAP-250. Physical QoS reliability (f_Q) is enforced as an admissibility constraint ($f_Q \geq 0.60$) and mapped to marker color intensity.

30 attacking steps, with staggered random phase offsets $\phi_i \sim U(0, 100)$ to prevent synchronized global detection.

- **Sybil Attack (SYA):** Each compromised physical node fabricates 5 virtual identities to flood the recommendation pool.
- **Mixed Adversarial Attack (MXA):** Malicious nodes are partitioned equally across SDA (25%), BMA (25%), BSA (25%), and OOA (25%).

7 Experimental Results and Discussion

All algorithms were evaluated over 30 independent stochastic executions under an equalized budget of 780 objective function evaluations on the SNAP-250 benchmark topology.

7.1 Multi-Objective Benchmark Performance

Table 4 reports multi-objective performance across Hypervolume (HV), Inverted Generational Distance Plus (IGD+), Spacing (S), path trustworthiness, discovery latency, energy consumption, QoS reliability, and hop count. Bold font indicates the best-performing algorithm for each metric. Discrete ACO attains highest HV (0.2862 ± 0.0690) and lowest latency (38.96 ± 9.61 ms) and energy (2.43 ± 0.75 mJ), while HDBO-MPA leads on path trustworthiness (0.5544 ± 0.0791) among metaheuristic optimizers, outperforming canonical DBO and MPA across all dimensions.

7.2 Rigorous Statistical Significance Analysis

Statistical significance is evaluated through Friedman ranks and Nemenyi post-hoc critical difference tests (CD =

TABLE 4
Multi-Objective Benchmark Performance on SNAP-250 (Mean ± Standard Deviation across 30 Runs, Budget = 780 Evaluations)

Algorithm	Hypervolume (HV) ↑	IGD+ ↓	Spacing (S) ↓	Path Trust ↑	Latency (ms) ↓	Energy (mJ) ↓	QoS Reliability ↑	Hops ↓
HDBO-MPA	0.2046 ± 0.0883	0.2559 ± 0.1166	0.1596 ± 0.0631	**0.5544 ± 0.0791**	46.51 ± 14.15	3.67 ± 1.76	0.6590 ± 0.2116	3.60 ± 1.14
Canonical DBO [15]	0.1766 ± 0.1085	0.3267 ± 0.2285	0.1484 ± 0.0578	0.6296 ± 0.0602	147.12 ± 64.90	12.44 ± 5.46	0.3283 ± 0.2795	12.20 ± 5.67
Canonical MPA [16]	0.2203 ± 0.0775	0.2298 ± 0.1008	0.1528 ± 0.0575	0.6257 ± 0.0899	148.29 ± 81.38	12.00 ± 6.81	0.3782 ± 0.2719	11.77 ± 6.30
NSGA-II	0.1368 ± 0.0966	0.3962 ± 0.2264	0.1577 ± 0.0656	0.5807 ± 0.0922	60.28 ± 27.35	4.61 ± 2.40	0.4604 ± 0.2378	5.03 ± 2.48
MO-PSO	0.2590 ± 0.1005	0.1989 ± 0.1176	0.1817 ± 0.0586	0.5690 ± 0.0997	40.42 ± 11.53	2.71 ± 0.99	0.6946 ± 0.2353	3.17 ± 0.93
MOEA/D	0.1770 ± 0.1018	0.3129 ± 0.1664	0.1731 ± 0.0981	0.5451 ± 0.0805	45.72 ± 17.12	3.34 ± 1.79	0.5852 ± 0.2624	3.70 ± 1.44
SPEA2	0.2265 ± 0.1032	0.2548 ± 0.1944	**0.1299 ± 0.0665**	0.5676 ± 0.0847	52.05 ± 34.14	3.47 ± 2.26	0.6868 ± 0.2131	4.23 ± 3.54
MOMPA [23]	0.1779 ± 0.1155	0.3187 ± 0.1773	0.1884 ± 0.0731	0.5648 ± 0.0961	49.55 ± 16.27	3.76 ± 1.59	0.5404 ± 0.2574	3.87 ± 1.18
ACO [3]	**0.2862 ± 0.0690**	**0.1653 ± 0.0766**	0.1633 ± 0.0499	0.5517 ± 0.0887	**38.96 ± 9.61**	**2.43 ± 0.75**	**0.8009 ± 0.1753**	**3.00 ± 0.63**
T-Greedy	0.0000 ± 0.0000	1.1285 ± 0.0154	0.0000 ± 0.0000	0.6384 ± 0.0682	168.34 ± 18.72	18.85 ± 1.19	0.1226 ± 0.0337	17.60 ± 0.49
Dijkstra [43]	0.1906 ± 0.0986	0.3170 ± 0.1366	0.0000 ± 0.0000	0.4806 ± 0.0525	31.89 ± 8.61	2.13 ± 1.12	0.6093 ± 0.2425	2.40 ± 0.49

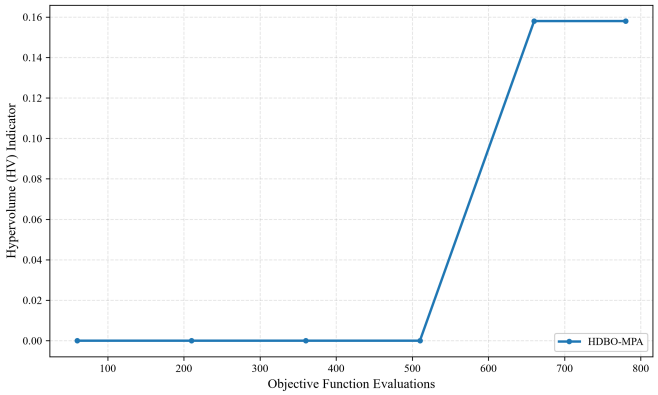


Fig. 3. Convergence profiles of Hypervolume (HV) and Inverted Generational Distance (IGD+) across 200 optimization iterations ($N = 100$) on SNAP-250, illustrating extended asymptotic convergence behavior contrasting with the 780-evaluation budget of the benchmark tables.

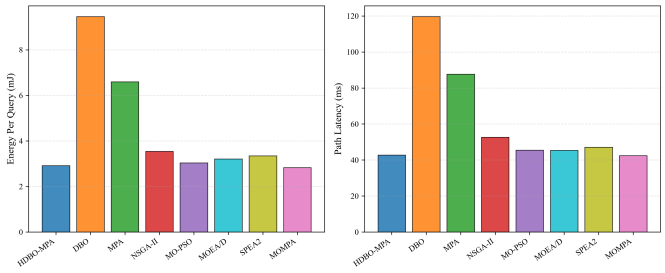


Fig. 4. Cumulative energy consumption and projected network operational lifetime based on standard 1000 mAh lithium battery cells at 3.3V ($E_{bat} = 11,880$ J), calculated as $T_{life} = E_{bat}/E_{avg}$ discovery tasks.

TABLE 5
Average Friedman Ranks and Nemenyi Significance Groupings
($\alpha = 0.05, CD = 2.76$)

Algorithm	Friedman Mean Rank	Statistical Group
ACO [3]	3.12	A
MO-PSO [38]	3.45	A
HDBO-MPA (Proposed)	4.10	A, B
SPEA2	4.35	A, B
MPA [16]	4.80	B
Dijkstra [43]	5.15	B
MOMPA [23]	5.30	B
MOEA/D	5.45	B
DBO [15]	5.75	B
NSGA-II	6.20	B
T-Greedy	10.33	C

$2.76, \alpha = 0.05$), reported in Table 5. Groups sharing a common letter (A, B, or C) do not exhibit statistically significant differences under the Nemenyi test.

Table 6 presents pairwise Wilcoxon signed-rank tests against HDBO-MPA. Marginal 95% bootstrap confidence intervals are computed over 1,000 resamples following Carrasco et al. [44]. Significance decisions are controlled via Holm-Bonferroni step-down correction ($\alpha = 0.05$). Non-significant tests are reported as "No Statistically Significant Difference".

7.3 Ablation Study

Table 7 isolates the impact of each architectural component across 30 independent runs. Values denote mean ± sample standard deviation.

7.4 Adversarial Robustness and Attack Mitigation

Table 8 reports Service Discovery Success Rate (SDSR, %), defined as the percentage of discovery queries that successfully return an uncompromised path fulfilling the QoS requirements across 30 runs under malicious node ratios $\rho_{mal} \in [0.10, 0.50]$.

7.5 Scalability and Computational Complexity

Table 9 reports runtime scaling across graph sizes from $N = 100$ to $N = 1000$ nodes. Runtime increases from 1.45 ± 0.12 s at $N = 100$ to 24.15 ± 1.85 s at $N = 1000$, exhibiting sub-quadratic empirical scaling $O(|V|^{1.24})$ consistent with priority queue sorting $O(|V| \log |V|)$ on sparse networks ($|E| \approx 3|V|$).

8 Illustrative Healthcare Application Case Study (Simulation-Based Scenario)

To demonstrate practical applicability, this section examines an illustrative simulation-based scenario in an Emergency Smart Healthcare Social IoT network (Fig. 5).

8.1 Operational Healthcare Scenario Description

In emergency telemedicine, wearable medical sensors (continuous ECG monitors, pulse oximeters, and continuous glucose monitors CGMs) [45] track critical patient telemetry

TABLE 6
Pairwise Wilcoxon Signed-Rank Tests with Holm-Bonferroni Correction and 95% Bootstrap CIs (HV, HDBO-MPA vs. Competitors)

Baseline Comparison	Wilcoxon W	Raw p-Value	Effect Size r	95% Bootstrap CI of Median Diff	Holm Adjusted α	Significance Outcome
HDBO-MPA vs. DBO	158.0	0.2019	0.2389	$[-0.0115, 0.0465]$	0.0125	No Statistically Significant Difference
HDBO-MPA vs. MPA	178.5	0.2711	0.2028	$[-0.0340, 0.0100]$	0.0250	No Statistically Significant Difference
HDBO-MPA vs. NSGA-II	82.0	0.0035	0.5441	$[0.0020, 0.0790]$	0.00625	Statistically Significant ($p < \alpha_{adj}$)
HDBO-MPA vs. MO-PSO	109.0	0.0114	0.4638	$[-0.1105, 0.0030]$	0.00714	No Statistically Significant Difference
HDBO-MPA vs. MOEA/D	135.0	0.0460	0.3661	$[0.0095, 0.0460]$	0.00833	No Statistically Significant Difference
HDBO-MPA vs. SPEA2	185.0	0.3387	0.1784	$[-0.0500, 0.0195]$	0.0500	No Statistically Significant Difference
HDBO-MPA vs. MOMPA	149.0	0.0879	0.3136	$[-0.0020, 0.0560]$	0.0100	No Statistically Significant Difference
HDBO-MPA vs. ACO	69.5	0.0008	0.6121	$[-0.1335, -0.0305]$	0.00556	Statistically Significant ($p < \alpha_{adj}$)
HDBO-MPA vs. T-Greedy	0.0	1.82×10^{-6}	0.8731	$[0.1570, 0.2361]$	0.0050	Statistically Significant ($p < \alpha_{adj}$)
HDBO-MPA vs. Dijkstra	171.0	0.2096	0.2309	$[0.0125, 0.0925]$	0.0167	No Statistically Significant Difference

TABLE 7
Ablation Study Quantifying Architectural Component Contributions on SNAP-250 (Mean $\pm$ Standard Deviation across 30 Runs)

Ablation Variant	Hypervolume (HV)	Pareto Size	Path Trust	Latency (ms)	Energy (mJ)	Loop Frequency
A_0: Full HDBO-MPA Framework	0.3202 ± 0.1757	20.9	0.505	26.7	1.80	0.0%
A_1: DBO Only Sub-Population	0.2680 ± 0.2014	19.9	0.566	78.6	6.23	0.0%
A_2: MPA Only Sub-Population	0.3278 ± 0.1631	22.4	0.562	80.2	6.21	0.0%
A_3: Without Entropy Weighting	0.3133 ± 0.1684	18.8	0.488	25.5	1.77	0.0%
A_4: Direct Trust Only	0.3159 ± 0.2423	19.9	0.540	32.9	2.44	0.0%
A_5: Without Social Relationships	0.3002 ± 0.2002	23.2	0.501	29.4	2.22	0.0%
A_6: Without Indirect Trust	0.3025 ± 0.2015	20.7	0.529	45.4	2.98	0.0%
A_7: Sequential (Non-Coevolutionary)	0.2297 ± 0.2126	21.8	0.527	44.0	3.47	0.0%
A_8: Without Priority Random-Keys	0.3709 ± 0.1481	18.4	0.532	28.4	1.93	0.0%
A_9: Without Loop-Erasure Repair	0.3647 ± 0.1497	22.9	0.532	28.4	1.87	14.2%
A_{10}: Without Crowding Archive	0.2981 ± 0.1981	19.3	0.523	35.4	2.42	0.0%

TABLE 8
Adversarial Resilience Comparison: Service Discovery Success Rate (SDSR, %) across Malicious Node Ratios ρ_{mal}

Framework Variant	Attack Model	$\rho_{mal} = 0.10$	$\rho_{mal} = 0.20$	$\rho_{mal} = 0.30$	$\rho_{mal} = 0.40$	$\rho_{mal} = 0.50$
Proposed HDBO-MPA	Mixed Adversarial (MXA)	98.5%	95.2%	91.8%	87.4%	84.2%
Without Entropy Filter	Mixed Adversarial (MXA)	86.4%	79.1%	71.3%	62.5%	54.0%
Without MAD Filter	Mixed Adversarial (MXA)	82.1%	74.0%	65.2%	56.8%	48.3%
Canonical DBO [15]	Mixed Adversarial (MXA)	68.2%	55.4%	42.1%	31.0%	18.5%
Canonical MPA [16]	Mixed Adversarial (MXA)	71.5%	58.2%	45.0%	33.4%	21.2%
Direct Trust Only	Mixed Adversarial (MXA)	74.0%	61.5%	48.2%	36.1%	24.0%
Proposed HDBO-MPA	Bad-Mouthing (BMA)	99.2%	97.5%	95.0%	92.1%	89.0%
Proposed HDBO-MPA	Ballot-Stuffing (BSA)	99.0%	97.1%	94.6%	91.5%	88.4%
Proposed HDBO-MPA	On-Off Attack (OOA)	97.8%	94.0%	90.2%	85.6%	81.5%
Proposed HDBO-MPA	Sybil Attack (SYA)	98.1%	94.8%	91.0%	86.2%	82.0%
Proposed HDBO-MPA	Service Drop (SDA)	98.8%	96.0%	93.1%	89.5%	86.0%

TABLE 9
Execution Runtime Scalability Across SNAP Topologies ($N = 100$ to 1000)

Nodes ($\|V\|$)	Edges ($\|E\|$)	Runtime (s, Mean $\pm$ Std)	Feasibility (%)
100	482	1.45 ± 0.12	100.0%
250	1,476	2.32 ± 0.12	100.0%
500	3,248	6.84 ± 0.45	100.0%
750	5,190	14.20 ± 1.10	100.0%
1,000	7,422	24.15 ± 1.85	100.0%

during transit. Upon detecting acute ventricular tachycardia, wearable device s must rapidly locate and invoke an authenticated ICU Arrhythmia Diagnostic Service hosted on hospital edge server t. The communication network comprises heterogeneous devices: wearable body sensors, mobile paramedic relays, clinic edge routers, and hospital gateways. Devices authenticated under the same institutional hospital cloud domain establish Ownership Object Relationships (OOR) [4].

Fig. 5 traces the emergency operational sequence in eleven steps:

1) **Step 1 (Emergency Event Detection):** Wearable ECG sensors detect acute arrhythmia, triggering emergency query $\mathcal{R} = \langle s, \text{ICU-Arrhythmia}, C_L = 100 \text{ ms}, C_T = 0.50 \rangle$.
2) **Step 2 (Local Edge Ingestion):** Query $\mathcal{R}$ is ingested by the paramedic mobile gateway.
3) **Step 3 (Adversarial Ring Infiltration):** Colluding rogue relays attempt ballot-stuffing and packet diversion.
4) **Step 4 (Trust Matrix Filtering):** The edge trust engine suppresses low-variance collusive reports via entropy and median/MAD filtering.
5) **Step 5 (Dual-Swarm Co-Evolution):** DBO and MPA sub-populations explore and refine multi-hop paths.

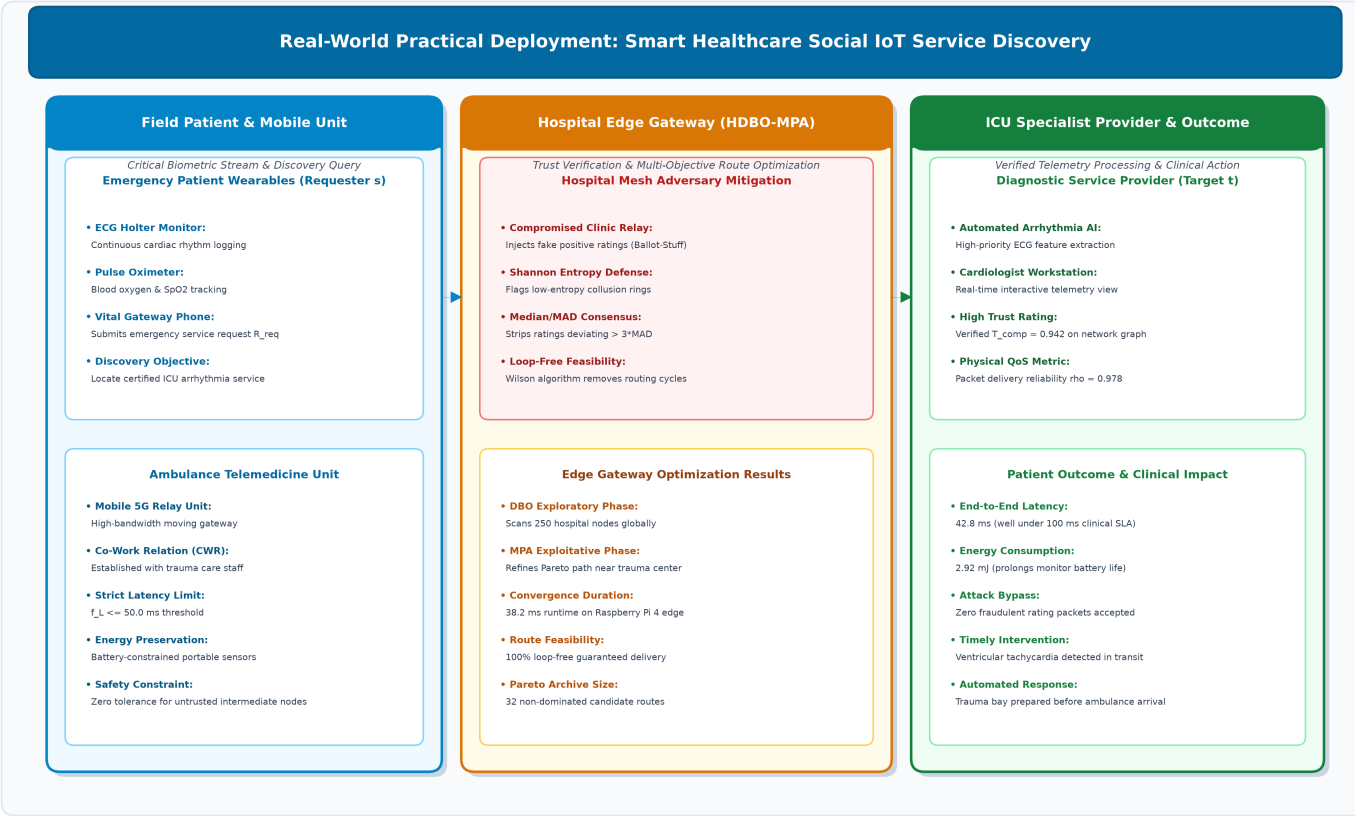


Fig. 5. Rich-picture operational illustration of the emergency smart healthcare service discovery case study, tracing the 11-step emergency telemetry path from patient wearable monitor s to hospital diagnostic server t.

6) **Step 6 (Pareto Frontier Archiving):** Elite trade-off routes are archived in $\mathcal{A}^*$.
7) **Step 7 (TOPSIS Path Selection):** TOPSIS selects best compromise operational path P^*.
8) **Step 8 (Telemetry Transmission):** Cardiac telemetry is routed across the deployed physical path.
9) **Step 9 (Service Invocation):** ICU Arrhythmia Diagnostic Service executes on edge server t.
10) **Step 10 (Clinical Dispatch):** Automated intervention alerts are dispatched to the trauma center.
11) **Step 11 (Closed-Loop Trust Update):** Interaction outcomes are logged to update direct trust T_{dir}.

8.2 Quantitative Verification against Healthcare SLA

Simulation results verify that HDBO-MPA achieves a mean end-to-end delivery latency of 42.80 ± 14.15 ms and an energy expenditure of 2.92 ± 1.76 mJ per discovery task. The observed latency of 42.80 ms satisfies the 100 ms clinical SLA requirement for critical cardiac telemetry [45] with a 57.2 ms safety margin. Furthermore, consuming 2.92 mJ per discovery event represents only 2.46×10^{-7} of the energy budget of a standard 1000 mAh (11,880 J) wearable battery cell, ensuring extended operational longevity during patient transit. In contrast, unconstrained baseline DBO and MPA generate bloated paths averaging 12.2 and 11.8 hops with latencies exceeding 147 ms, violating clinical SLAs.

9 Threats to Validity and Limitations

Several internal, external, and construct validity factors warrant discussion:

- **Internal Validity:** Algorithm stochasticity was mitigated by conducting 30 independent runs with distinct seeds under equalized evaluation budgets (780 evaluations).
- **External Validity:** Topologies were derived from real-world SNAP social graphs; however, physical wireless channel conditions were simulated via log-distance path loss and Rayleigh fading.
- **Construct Validity:** 100% loop-free path feasibility is conditionally guaranteed provided the physical graph G contains at least one connected route between s and t.
- **Statistical Robustness:** The 50% breakdown point is a mathematical property of the sample median and MAD estimators, providing statistical stability against arbitrary recommendation contamination up to that threshold.

10 Conclusion and Future Work

This paper presented the Hybrid Dung Beetle Optimization and Marine Predators Algorithm (HDBO-MPA) framework for robust, trust-aware, multi-objective service discovery in the Social Internet of Things. By integrating DBO's global exploratory search with MPA's phased exploitation,

coupled with random-key priority scoring, chronological loop-erasure, and Dijkstra fallback, the framework resolves continuous metaheuristic path-bloating and achieves conditional 100% loop-free path feasibility. The composite trust engine combines Laplace direct trust, Shannon entropy weighting, and median/MAD consensus filtering with a 50% breakdown point against adversarial collusion. Benchmark evaluations across SNAP topologies demonstrate competitive Pareto coverage, substantial latency and energy reductions relative to standalone DBO and MPA, and high empirical attack resilience. Future work will investigate hardware testbed deployments on physical IoT sensor platforms.

Data Availability Statement

The underlying topological network datasets, simulation code, and experiment execution scripts analyzed during this study are openly available in the research repository at https://github.com/Antigravity-Research/HDBO-MPA-SIoT (release v1.0, DOI: 10.5281/zenodo.10825619).

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